SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Reuirement Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 1 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:	Manish Kumar	**Reg. No.:**	1924260185
Date:	14/10/2026	**Duration:**	60 minutes

Code of Conduct

*I **MANISH KUMAR / 1924260185**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: manish kumar

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description
   - Scope of the System

- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

INTELLIGENT DIGITAL TWIN MONITORING PLATFORM FOR INDUSTRIAL ASSETS

1. Study and Analyze the Scenario

Industry Domain: Industrial asset monitoring and maintenance, covering turbines, pumps, compressors, and production systems.

Problem Description: Industrial organizations need continuous visibility into complex assets. Asset information is often distributed across sensor readings, operational records, performance data, and maintenance history. The proposed platform maintains a digital representation of each physical asset and synchronizes these information sources to support monitoring and decision-making.

Stakeholders: Plant operators, maintenance engineers, reliability engineers, plant managers, system administrators, and sensor/industrial control system integrators.

Objectives: Create digital asset models; synchronize sensor data; monitor equipment status; maintain maintenance history; visualize performance; and identify abnormal conditions.

2. Scope of the System

The platform will register industrial assets, maintain their digital-twin profiles, ingest and time-stamp sensor data, display current status and historical trends, record maintenance activities, detect abnormal readings using configurable rules, and provide dashboards and alerts. It will support human review and maintenance planning. Automatic control of physical machinery is outside the initial scope.

3. Functional Requirements

- FR-1: The system shall allow authorized users to create, view, update, and deactivate digital asset records.
- FR-2: Each asset record shall store an asset ID, type, location, specifications, operating limits, and associated sensors.
- FR-3: The system shall receive sensor readings from configured data sources and associate each reading with the correct asset and timestamp.
- FR-4: The system shall validate incoming readings and flag missing, invalid, or stale data.
- FR-5: The system shall display asset status, key measurements, trends, and alerts on a monitoring dashboard.
- FR-6: The system shall store and retrieve maintenance history, including work performed, date, findings, and responsible personnel.
- FR-7: The system shall compare incoming readings with configured thresholds or rules and identify abnormal conditions.
- FR-8: The system shall notify authorized users when configured abnormal conditions or data-quality issues occur.
- FR-9: The system shall provide asset-level historical views and performance summaries for a selected time period.
- FR-10: The system shall support role-based access and maintain an audit trail of important user actions.

4. Non-Functional Requirements

- Performance: Dashboard updates and routine queries should respond promptly under the expected plant workload; target values shall be agreed during deployment.
- Reliability: Sensor ingestion and stored records should be protected against accidental loss, with retry and recovery mechanisms.
- Availability: The platform should remain available during agreed operating hours and recover gracefully from component failures.
- Security: Authenticate users, apply role-based permissions, encrypt data in transit, and protect credentials and stored operational information.
- Scalability: Support adding assets, sensors, and sites without redesigning the whole platform.
- Usability: Present status, trends, alerts, and maintenance records in clear, accessible dashboards.

- Maintainability: Use documented interfaces, configurable thresholds, logging, and modular components.
- Data integrity: Store timestamps, units, source identifiers, and validation status consistently.

5. Actors and Use Cases

Primary actors: Plant Operator – views asset status and alerts; Maintenance Engineer – reviews abnormal conditions and records maintenance; Reliability Engineer – analyzes trends and configures limits; Plant Manager – reviews performance summaries; System Administrator – manages users, roles, and integrations; Sensor/Data Gateway – supplies readings.

Major use cases: Manage asset profile; ingest sensor data; view live status; view performance trends; detect and review anomalies; acknowledge alerts; record maintenance; view maintenance history; generate performance summaries; configure thresholds; manage users and permissions.

Use Case Diagram (text representation):

Sensor/Data Gateway → Submit Sensor Readings → Digital Twin Monitoring Platform

Plant Operator → View Asset Status / Review Alerts

Maintenance Engineer → Review Alerts / Record Maintenance / View History

Reliability Engineer → Analyze Trends / Configure Thresholds

Plant Manager → View Performance Dashboard / Generate Summary

System Administrator → Manage Assets, Users, Roles, and Integrations

6. Use Case Descriptions

Use Case: Monitor Asset. Actor: Plant Operator. Preconditions: Asset and sensors are registered. Main flow: user selects an asset; system retrieves the latest valid readings and status; dashboard shows measurements, trends, and active alerts. Alternative flow: if data is stale or unavailable, the system clearly marks the data condition. Postcondition: operator can review the current reported state.

Use Case: Detect Abnormal Condition. Actor: Monitoring service, with Reliability Engineer/Plant Operator as reviewers. Preconditions: sensor mapping and rules are configured. Main flow: system validates a reading; compares it with configured limits or rules; creates an event when a condition is met; records time, asset, measurement, and rule; notifies authorized users. Postcondition: an anomaly is available for review and acknowledgement.

Use Case: Record Maintenance. Actor: Maintenance Engineer. Preconditions: user is authenticated and the asset exists. Main flow: user selects an asset; enters maintenance date, work performed, findings, and notes; system validates and saves the entry; history is updated. Postcondition: the maintenance record is linked to the asset.

7. Data Requirements

Core entities: Asset (asset ID, name, type, location, specifications, operating limits); Sensor (sensor ID, asset ID, measurement type, unit, source); SensorReading (reading ID, sensor ID, timestamp, value, quality/status); Alert/Event (event ID, asset ID, rule, severity, timestamp, state, acknowledgement); MaintenanceRecord (record ID, asset ID, date, work description, findings, personnel); User/Role (user ID, role, permissions); and AuditLog (action, actor, timestamp, outcome). Data should use consistent units, timestamps, identifiers, access controls, retention rules, and backup policies.

8. External Interface Requirements

The platform shall provide a web-based dashboard for users and documented interfaces for sensor gateways or industrial data sources. Interfaces should support secure communication, authenticated access, timestamped readings, error reporting, and configuration of asset/sensor mappings. Specific protocols and integrations will be selected based on the existing plant environment.

9. System Features

Digital asset registry; sensor-data synchronization; real-time and historical monitoring; maintenance-history management; performance visualization; configurable anomaly detection; alert acknowledgement; role-based access; and audit logging.

10. Assumptions and Constraints

Assumptions: Assets have identifiable IDs; suitable sensors or data gateways are available; authorized personnel maintain asset and maintenance records; operating limits can be configured by qualified staff; and users have access to the required network and devices.
Constraints: Sensor compatibility and network connectivity may vary by site; data quality depends on sensor condition and calibration; storage and processing capacity depend on deployment resources; security and industrial safety policies must be followed; project cost and schedule may limit integrations; and the initial system provides monitoring and decision support rather than direct machine control.

11. Requirement Validation and Consistency

The requirements cover all stated objectives: digital asset models (FR-1/FR-2), sensor synchronization (FR-3/FR-4), status monitoring (FR-5), maintenance history (FR-6), performance visualization (FR-9), and abnormal-condition identification (FR-7/FR-8). Functional and non-functional requirements are separated, actors have corresponding use cases, and core data entities support the required features. During stakeholder review, confirm update frequency, alert severity levels, retention period, availability target, access roles, and exact interface protocols. These values are deployment-specific and should be agreed rather than assumed.

○

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5

Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25